

Physics

40 supplementary questions

Coverage expansion set

Original multiple-choice questions to supplement Mocks A-E. The topics are deliberately weighted towards gaps identified in that set.

Suggested time: 60 minutes for all 40 questions.

Choose one answer, A-E, for each question. Do not use a calculator. Diagrams are not to scale unless indicated. All necessary numerical constants are given in the relevant question.

This is a targeted question bank, not a full 27-question mock. It can also be attempted as two sets of 20 questions in approximately 30 minutes each.

Answer record

01 _____	11 _____	21 _____	31 _____
02 _____	12 _____	22 _____	32 _____
03 _____	13 _____	23 _____	33 _____
04 _____	14 _____	24 _____	34 _____
05 _____	15 _____	25 _____	35 _____
06 _____	16 _____	26 _____	36 _____
07 _____	17 _____	27 _____	37 _____
08 _____	18 _____	28 _____	38 _____
09 _____	19 _____	29 _____	39 _____
10 _____	20 _____	30 _____	40 _____

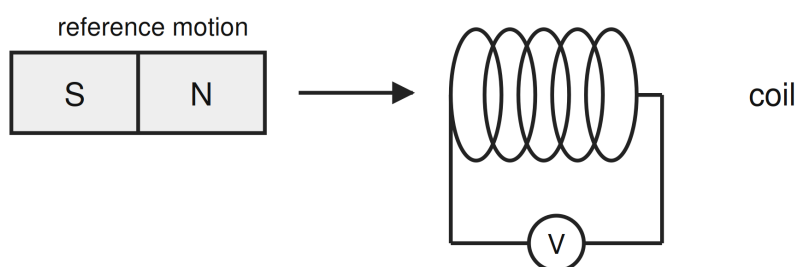
Independent practice material. Not an official UAT-UK, Cambridge or Pearson paper.

1

A magnet is moved along the axis of a stationary coil. Moving its north pole towards the coil gives a positive voltage at the voltmeter.

In a second experiment, its south pole is moved towards the coil, held stationary for a short time, and then moved away. The voltmeter connections are unchanged.

Which sequence describes the signs of the voltages during these three stages?

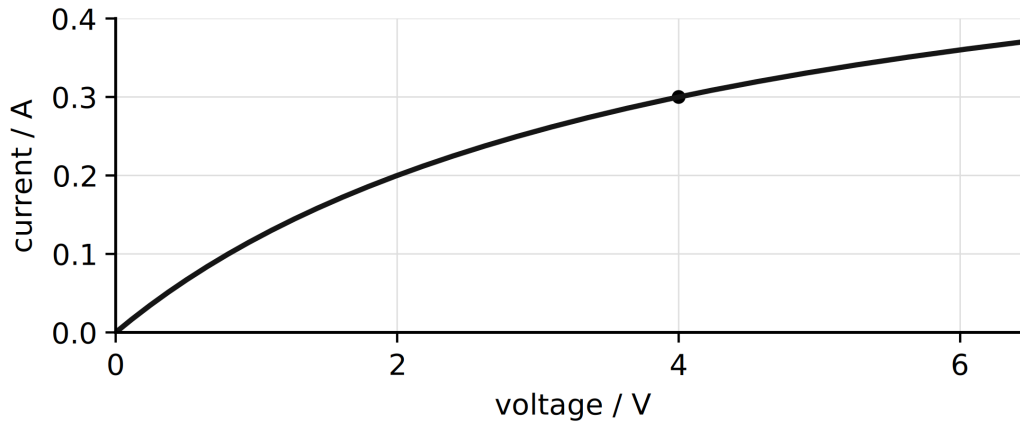


- A** negative, zero, positive
- B** positive, zero, negative
- C** negative, negative, positive
- D** positive, zero, positive
- E** negative, zero, negative

2

The graph shows the current through a filament lamp at different voltages. The lamp is connected in series with a $10\ \Omega$ resistor to a $7.0\ \text{V}$ d.c. supply.

What is the power transferred to the lamp?



- A** 0.90 W
- B** 1.2 W
- C** 1.6 W
- D** 2.1 W
- E** 2.8 W

3

A spring obeys Hooke's law. A force of $6.0\ \text{N}$ produces an extension of $3.0\ \text{cm}$.

The spring is already extended by $2.0\ \text{cm}$. How much additional energy must be transferred to the spring to increase its extension to $5.0\ \text{cm}$?

- A** 0.09 J
- B** 0.15 J
- C** 0.21 J
- D** 0.25 J
- E** 0.29 J

4

A detector records the mean count rates shown for a source emitting two types of radiation. Its background count rate is $20 \text{ counts min}^{-1}$.

The paper stops all alpha particles. The aluminium stops all alpha and beta particles. Neither absorber significantly affects gamma radiation.

Which two types are detected from the source, and what is the count rate due to the more penetrating type alone?

Absorber	Mean count rate / counts min^{-1}
none	240
paper	240
aluminium	90

- A** alpha and beta; $70 \text{ counts min}^{-1}$
- B** alpha and gamma; $90 \text{ counts min}^{-1}$
- C** beta and gamma; $70 \text{ counts min}^{-1}$
- D** beta and gamma; $90 \text{ counts min}^{-1}$
- E** beta and gamma; $150 \text{ counts min}^{-1}$

5

An ideal transformer has 800 turns on its primary coil and 40 on its secondary coil. The primary is supplied at 240 V a.c.

Identical lamps rated at 12 V, 6.0 W are connected in parallel across the secondary. The primary current must not exceed 0.15 A.

What is the largest number of lamps that can operate at their rated power?

- A** 3
- B** 4
- C** 5
- D** 6
- E** 12

6

Paint droplets are given a negative charge before being sprayed towards a positively charged metal panel. The droplets stick to the panel and transfer their excess electrons to it.

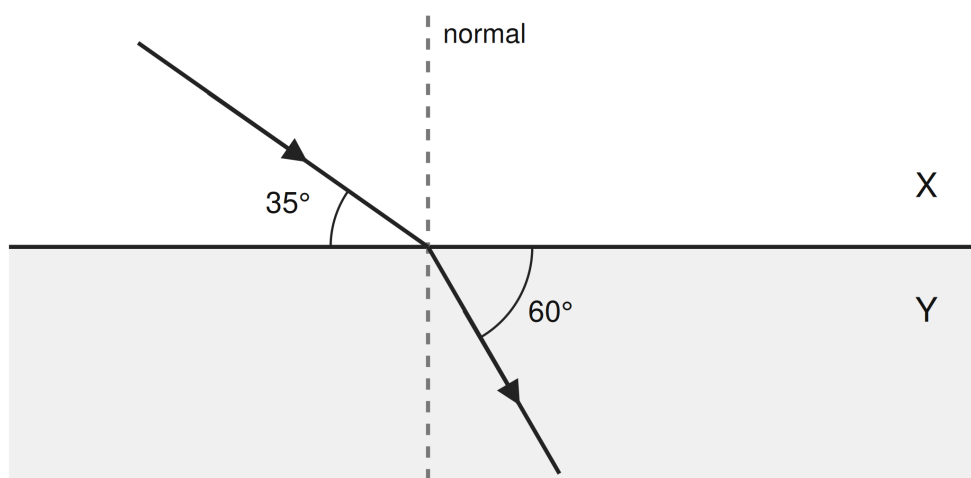
The panel is kept at a constant positive charge by an electrical connection. Which row describes the electric force between two droplets and the electron transfer needed through the connection?

- A** droplets attract; electrons enter the panel
- B** droplets attract; electrons leave the panel
- C** droplets repel; electrons enter the panel
- D** no force between droplets; no electron transfer
- E** droplets repel; electrons leave the panel

7

A light ray crosses a boundary from material X into material Y. The angles marked on the diagram are measured between the ray and the boundary.

Which statement correctly compares the wave in Y with the wave in X?

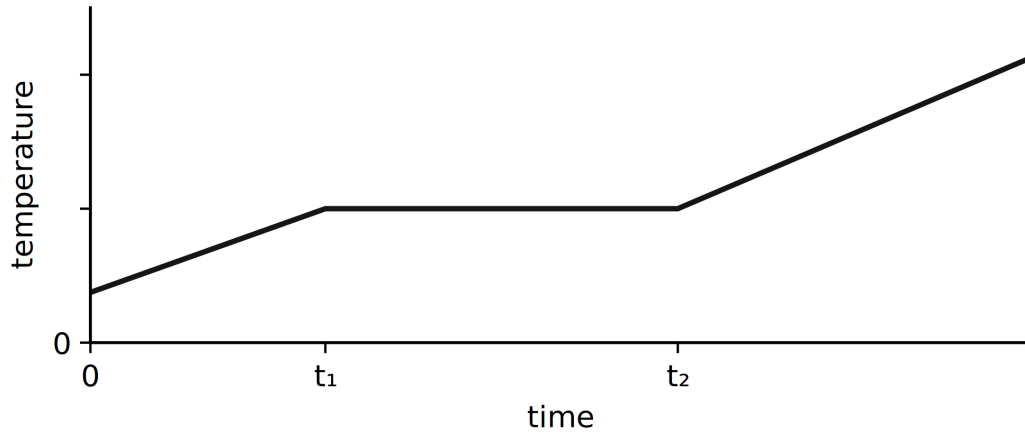


- A** lower speed, lower frequency
- B** lower speed, same frequency
- C** higher speed, same frequency
- D** higher speed, higher frequency
- E** same speed, same frequency

8

A pure substance is heated at constant power. The graph shows its temperature. During the horizontal section, solid and liquid are both present. Energy losses are negligible.

Which statement is correct during this horizontal section?

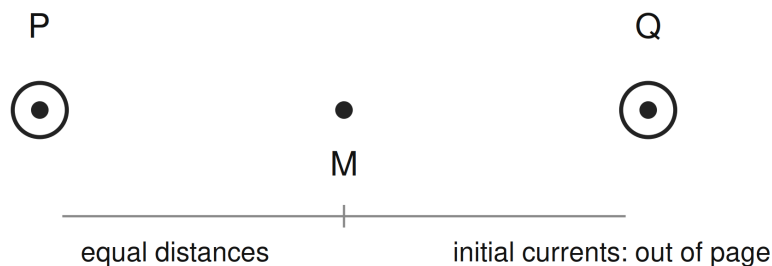


- A** The average kinetic energy of its particles stays constant while their potential energy increases.
- B** The average kinetic energy of its particles increases while their potential energy stays constant.
- C** No energy is transferred to the substance.
- D** Both the average kinetic energy and potential energy of its particles decrease.
- E** The particles stop moving until melting is complete.

9

Two long straight wires P and Q carry equal currents out of the page. Point M is midway between them. The magnetic field at M due to either wire alone has magnitude B . Other magnetic fields are negligible.

The current in Q is reversed without changing its magnitude. What is the resultant magnetic field at M?



- A** zero
- B** B , upwards on the page
- C** B , downwards on the page
- D** $2B$, downwards on the page
- E** $2B$, upwards on the page

10

An ultrasound pulse enters a layer of tissue along a line perpendicular to its two parallel surfaces. Echoes from the front and back surfaces return to the transmitter $30\ \mu\text{s}$ and $70\ \mu\text{s}$ after emission.

The speed of ultrasound in the layer is $1500\ \text{m s}^{-1}$. What is the thickness of the layer?

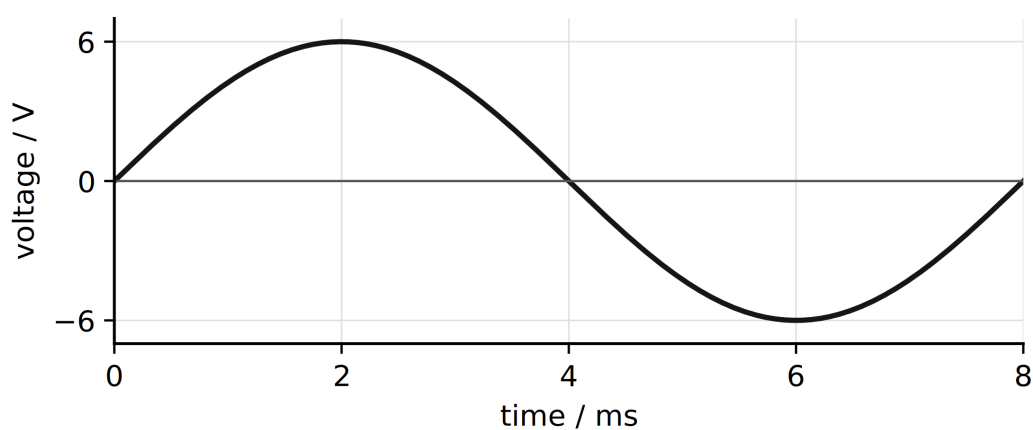
- A** 1.5 cm
- B** 3.0 cm
- C** 4.5 cm
- D** 6.0 cm
- E** 7.5 cm

11

The graph shows the voltage from a simple a.c. generator whose coil rotates at constant speed. One voltage cycle corresponds to one complete rotation.

The rotation speed is doubled and the magnetic field strength is left unchanged. The coil starts in the same position and rotates in the same direction.

What is the first time after $t = 0$ at which the new voltage reaches a positive maximum, and how does the new peak voltage compare with the old peak voltage?

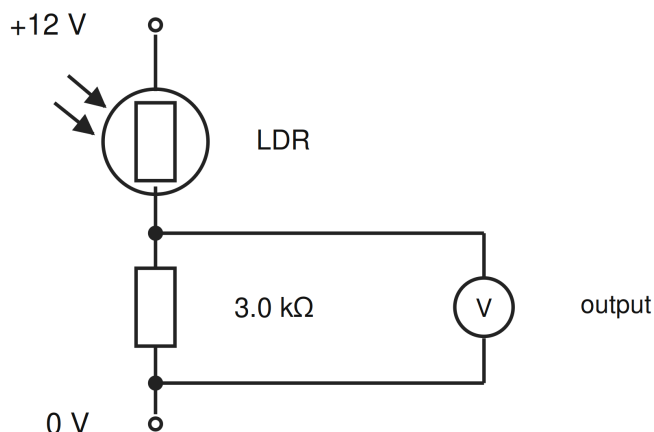


- A** 1.0 ms; smaller
- B** 1.0 ms; larger
- C** 2.0 ms; unchanged
- D** 4.0 ms; larger
- E** 8.0 ms; unchanged

12

An LDR and a $3.0\text{ k}\Omega$ resistor are connected in series across a 12 V supply, as shown. A control unit switches on when the voltage across the fixed resistor exceeds 8.0 V . The control unit draws negligible current.

At what LDR resistance is the voltage exactly 8.0 V , and which change will then switch the control unit on?



- A** $1.5\text{ k}\Omega$; increasing light intensity
- B** $1.5\text{ k}\Omega$; decreasing light intensity
- C** $2.0\text{ k}\Omega$; increasing light intensity
- D** $6.0\text{ k}\Omega$; increasing light intensity
- E** $6.0\text{ k}\Omega$; decreasing light intensity

13

A transmission line receives 20 kW at 2.0 kV . The total resistance of its cables is $5.0\text{ }\Omega$.

The voltage at the sending end is increased to 10 kV , while the power entering the line remains 20 kW . The cable resistance is unchanged.

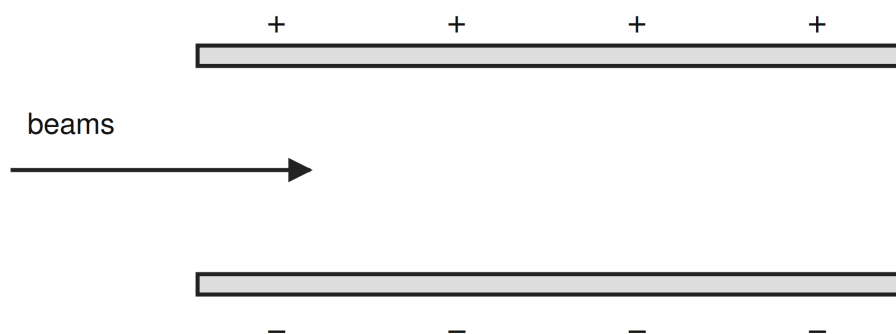
By how much does the power lost in the cables decrease?

- A** 20 W
- B** 100 W
- C** 400 W
- D** 480 W
- E** 500 W

14

Narrow beams of alpha particles, beta-minus particles and gamma radiation enter the region between two charged plates. The beams initially travel horizontally to the right. The upper plate is positive and the lower plate is negative.

Which row gives the direction in which each beam bends?

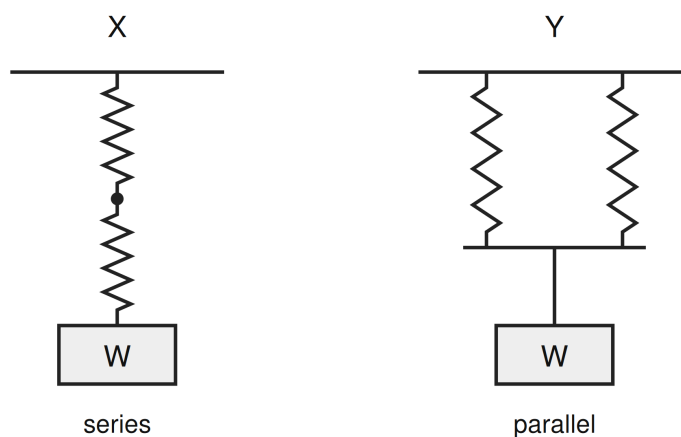


- A** alpha downwards; beta upwards; gamma straight
- B** alpha upwards; beta downwards; gamma straight
- C** alpha downwards; beta straight; gamma upwards
- D** alpha upwards; beta upwards; gamma straight
- E** alpha straight; beta upwards; gamma downwards

15

Two identical light springs obey Hooke's law. In arrangement X they are in series; in arrangement Y they are in parallel. The same load of weight W is supported at rest in each arrangement.

What is the ratio of the total elastic energy stored in X to that stored in Y?



- A** 1 : 4
- B** 1 : 2
- C** 1 : 1
- D** 2 : 1
- E** 4 : 1

16

Three devices use different regions of the electromagnetic spectrum:

1. a thermal camera detecting radiation emitted by a warm hand;
2. a lamp producing ultraviolet radiation;
3. a scanner producing X-rays.

Which row lists the devices in order of increasing radiation frequency and correctly compares the speeds of their radiation in a vacuum?

- A** 1, 2, 3; all three speeds are equal
- B** 1, 3, 2; all three speeds are equal
- C** 3, 2, 1; all three speeds are equal
- D** 1, 2, 3; speed increases in this order
- E** 2, 1, 3; speed decreases in this order

17

An initially uncharged insulating object loses 5.0×10^{10} electrons when rubbed. It later gains 2.0×10^{10} electrons.

What is its final charge? The magnitude of the charge on one electron is 1.6×10^{-19} C.

- A** -11.2 nC
- B** -4.8 nC
- C** $+3.2 \text{ nC}$
- D** $+4.8 \text{ nC}$
- E** $+11.2 \text{ nC}$

18

Straight water waves cross a boundary from deep water into shallow water. In the deep water, the distance from the first crest to the fifth crest is 24 cm. The wave speed there is 30 cm s^{-1} .

In the shallow water, the distance from the first crest to the fifth crest is 16 cm.

What are the wave frequency and the speed in the shallow water?

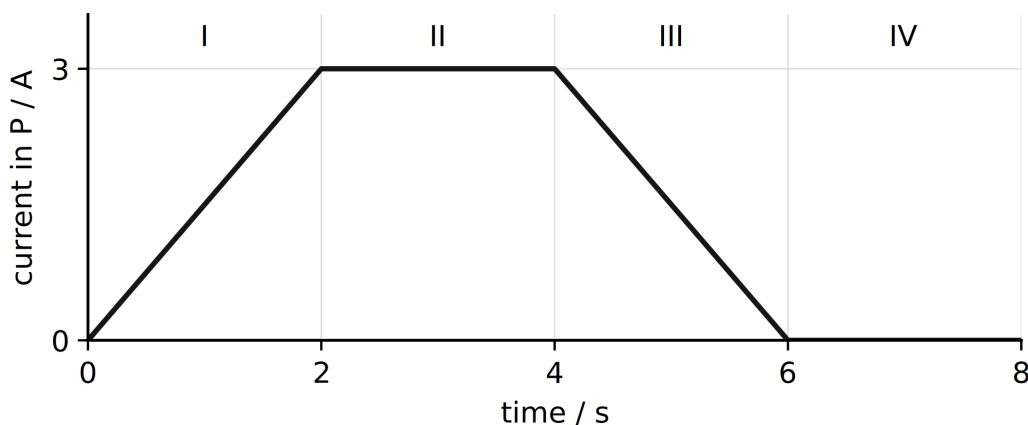
- A** $1.25 \text{ Hz}, 20 \text{ cm s}^{-1}$
- B** $5.0 \text{ Hz}, 20 \text{ cm s}^{-1}$
- C** $5.0 \text{ Hz}, 45 \text{ cm s}^{-1}$
- D** $7.5 \text{ Hz}, 30 \text{ cm s}^{-1}$
- E** $1.25 \text{ Hz}, 5.0 \text{ cm s}^{-1}$

19

Two stationary coils share an iron core. The graph shows the current in coil P. A voltmeter is connected across coil Q.

Increasing the current in P produces a positive voltage in Q. The voltmeter connections are fixed.

During which labelled intervals is the voltage in Q non-zero?



- A** I and II only
- B** all four intervals
- C** I, III and IV only
- D** II and IV only
- E** I and III only

20

An NTC thermistor is connected in series with a $2.0\text{ k}\Omega$ resistor across a constant 12 V supply. At the initial temperature the thermistor resistance is $4.0\text{ k}\Omega$.

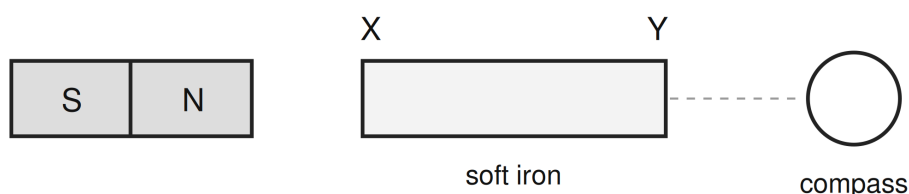
The thermistor is warmed until its resistance becomes $1.0\text{ k}\Omega$. What happens to the electrical power transferred to the thermistor?

- A** It decreases to one quarter of its initial value.
- B** It decreases to one half of its initial value.
- C** It remains unchanged.
- D** It doubles.
- E** It quadruples.

21

The north pole of a permanent magnet is brought close to end X of an initially unmagnetised soft-iron bar. The far end of the bar is Y.

A small compass is placed just beyond Y, on the axis of the bar. Which way does the north-seeking end of the compass point, and why is soft iron suitable for an electromagnet that must release its load when switched off?



- A** towards Y; it is difficult to magnetise
- B** towards Y; it retains a strong magnetisation
- C** away from Y; it retains a strong magnetisation
- D** towards Y; it loses most of its magnetisation when the applied field is removed
- E** away from Y; it loses most of its magnetisation when the applied field is removed

22

A shallow dish of water is left in a room at constant temperature. For a short time, evaporation cools the remaining water below room temperature.

Which pair of statements explains this cooling?

- A** Faster particles escape preferentially; the average kinetic energy of the remaining particles decreases.
- B** Slower particles escape preferentially; the average kinetic energy of the remaining particles decreases.
- C** Faster particles escape preferentially; the remaining particles become smaller.
- D** All particles slow down by the same amount before any can escape.
- E** Evaporation can occur only after every particle has the same kinetic energy.

23

An ideal transformer has one primary coil and two separate secondary coils. The primary has 600 turns and is supplied at 120 V a.c.

Secondary X has 60 turns and supplies a $6.0\ \Omega$ resistor. Secondary Y has 120 turns and supplies a $24\ \Omega$ resistor. Both loads are connected simultaneously.

What is the primary current?

- A** 0.20 A
- B** 0.30 A
- C** 0.40 A
- D** 0.60 A
- E** 3.0 A

24

An oscilloscope displays the signal from a microphone. For each sound, the trace measurements and oscilloscope settings are given in the table. The microphone's voltage amplitude is proportional to the sound-wave amplitude.

Compared with sound P, sound Q has:

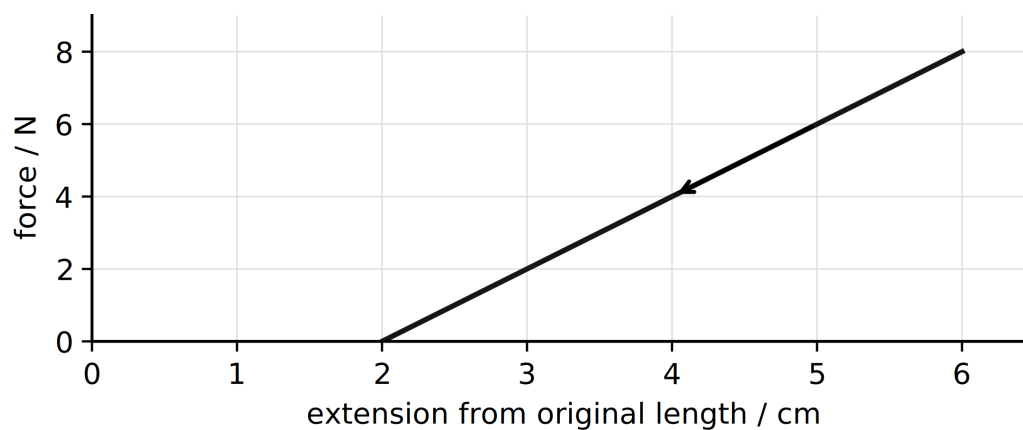
Sound	Divisions per cycle	Time / ms per division	Peak height / divisions	Voltage / V per division
P	4.0	0.50	3.0	0.20
Q	2.0	0.50	2.0	0.30

- A** half the frequency and the same amplitude
- B** the same frequency and half the amplitude
- C** the same frequency and the same amplitude
- D** double the frequency and double the amplitude
- E** double the frequency and the same amplitude

25

A spring has an original length of 10.0 cm. It is stretched beyond its elastic limit. The graph shows the force as the spring is then unloaded.

What is the spring's length when the force has fallen to 2.0 N, and what is its permanent increase in length after complete unloading?



- A** 11.0 cm; 2.0 cm
- B** 13.0 cm; zero
- C** 13.0 cm; 2.0 cm
- D** 14.0 cm; 2.0 cm
- E** 16.0 cm; 6.0 cm

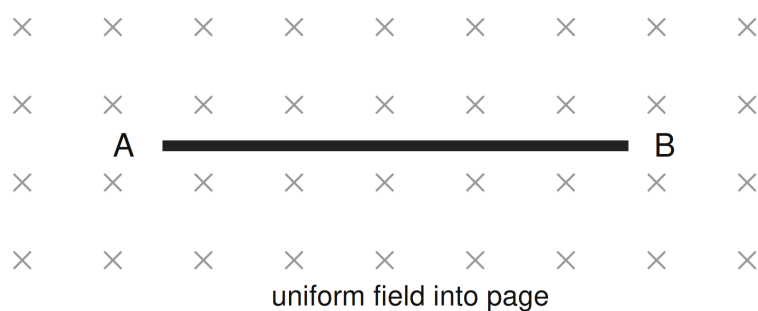
26

A straight horizontal metal rod lies in the plane of the page in a uniform magnetic field directed into the page. The rod is moved without rotating.

Consider these motions:

1. vertically upwards in the plane of the page;
2. horizontally to the right, along the rod's own length;
3. directly into the page, parallel to the magnetic field.

For which motions is a voltage induced between the two ends of the rod?

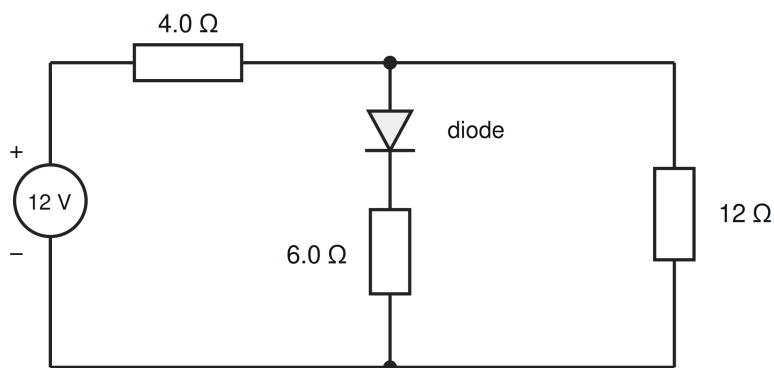


- A** 1 only
B 2 only
C 3 only
D 1 and 2 only
E all three

27

A 12 V d.c. supply is connected to the circuit shown. The ideal diode initially conducts. A conducting ideal diode has zero resistance; in the reverse direction it blocks all current.

The supply connections are reversed. What is the new total power supplied to the circuit?



- A** 6.0 W
- B** 9.0 W
- C** 12 W
- D** 18 W
- E** 36 W

28

A short pulse of light travels normally through a transparent block of thickness 12 cm. Its speed in the block is $2.0 \times 10^8 \text{ m s}^{-1}$.

The block is replaced by a second block of the same thickness in which the pulse travels at $1.5 \times 10^8 \text{ m s}^{-1}$. The rest of the pulse's path is unchanged.

By how much does its arrival at a detector become later?

- A** 0.10 ns
- B** 0.20 ns
- C** 0.40 ns
- D** 0.60 ns
- E** 0.80 ns

29

A source and detector monitor the thickness of a continuous sheet of paper. The source must work for several years without a large correction for radioactive decay. Alpha radiation is completely stopped by the paper; beta radiation is partly absorbed; gamma radiation is hardly affected by changes in the paper's thickness.

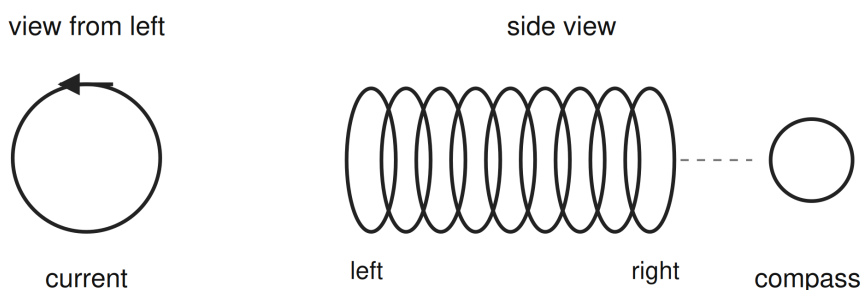
Which source is most suitable, and what happens to the detector count rate if the sheet becomes thinner?

- A** alpha, half-life 20 years; decreases
- B** beta, half-life 2 hours; increases
- C** beta, half-life 20 years; increases
- D** beta, half-life 20 years; decreases
- E** gamma, half-life 20 years; increases strongly

30

The left-hand end of a solenoid is viewed directly. The current around this end flows anticlockwise, as shown. A small compass is placed on the solenoid's axis just beyond its right-hand end. Other magnetic fields are negligible.

Which way does the compass's north-seeking end point initially, and after the current is reversed?



- A** right initially; right afterwards
- B** right initially; left afterwards
- C** left initially; left afterwards
- D** left initially; right afterwards
- E** upwards initially; downwards afterwards

31

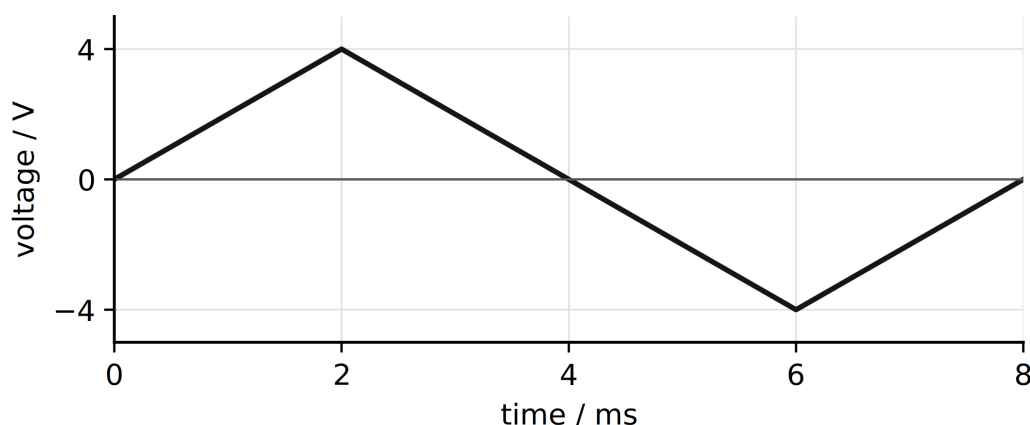
A fixed mass of gas is sealed in a rigid container and heated. Which statement best explains the increase in pressure?

- A** The particles have greater average speed and transfer more momentum to the walls per unit time.
- B** The particles become larger, so less empty space remains.
- C** The number of particles increases while their average speed remains constant.
- D** The particles move more slowly and therefore remain in contact with the walls for longer.
- E** The average spacing between particles must increase because the gas is hotter.

32

The graph shows the voltage produced by an electromagnetic generator during one complete cycle. It is connected in series with a $2.0\text{ k}\Omega$ resistor and an ideal diode. The diode allows current only when the plotted voltage is positive. The voltage is unchanged by the connection.

How much charge passes through the resistor during one complete cycle?



- A** $2.0\text{ }\mu\text{C}$
- B** $4.0\text{ }\mu\text{C}$
- C** $8.0\text{ }\mu\text{C}$
- D** $16\text{ }\mu\text{C}$
- E** zero

33

An ideal transformer supplies a fixed resistor. The primary voltage is held constant.

The transformer is replaced with another ideal transformer having half as many primary turns and the same number of secondary turns. The same resistor is connected across the secondary.

By what factor does the primary current change?

- A** $\frac{1}{4}$
- B** $\frac{1}{2}$
- C** 1
- D** 2
- E** 4

34

An insulated metal tank acquires a positive charge as fuel flows out. It is then connected to Earth by a conducting wire and becomes uncharged.

Which row correctly describes the charging process and the particle movement through the wire during discharge?

- A** The tank gained protons; protons move from the tank to Earth.
- B** The tank lost electrons; electrons move from the tank to Earth.
- C** The tank gained electrons; electrons move from Earth to the tank.
- D** The tank lost protons; protons move from Earth to the tank.
- E** The tank lost electrons; electrons move from Earth to the tank.

35

A station sends a microwave signal to a satellite 36000 km away. The satellite immediately sends an infrared signal to a second station 45000 km away from it.

Treat both paths as vacuum and ignore processing time. The speed of light is $3.0 \times 10^8 \text{ m s}^{-1}$.

What is the shortest time between emission at the first station and reception at the second?

- A** 0.030 s
- B** 0.12 s
- C** 0.15 s
- D** 0.27 s
- E** 0.54 s

36

A 0.050 kg block is launched along a horizontal track by a spring of spring constant 800 N m^{-1} , initially compressed by 5.0 cm. The spring obeys Hooke's law and is not attached to the block.

The block loses 0.20 J of energy on a rough section, then rises up a smooth slope. All other energy losses are negligible.

What maximum vertical height does it reach above its starting level? Take $g = 10 \text{ N kg}^{-1}$.

- A** 0.40 m
- B** 0.80 m
- C** 1.0 m
- D** 1.6 m
- E** 2.0 m

37

A sample initially contains 120 unstable nuclei of one isotope. Each nucleus decays directly to a stable nucleus. After one half-life, 72 unstable nuclei remain.

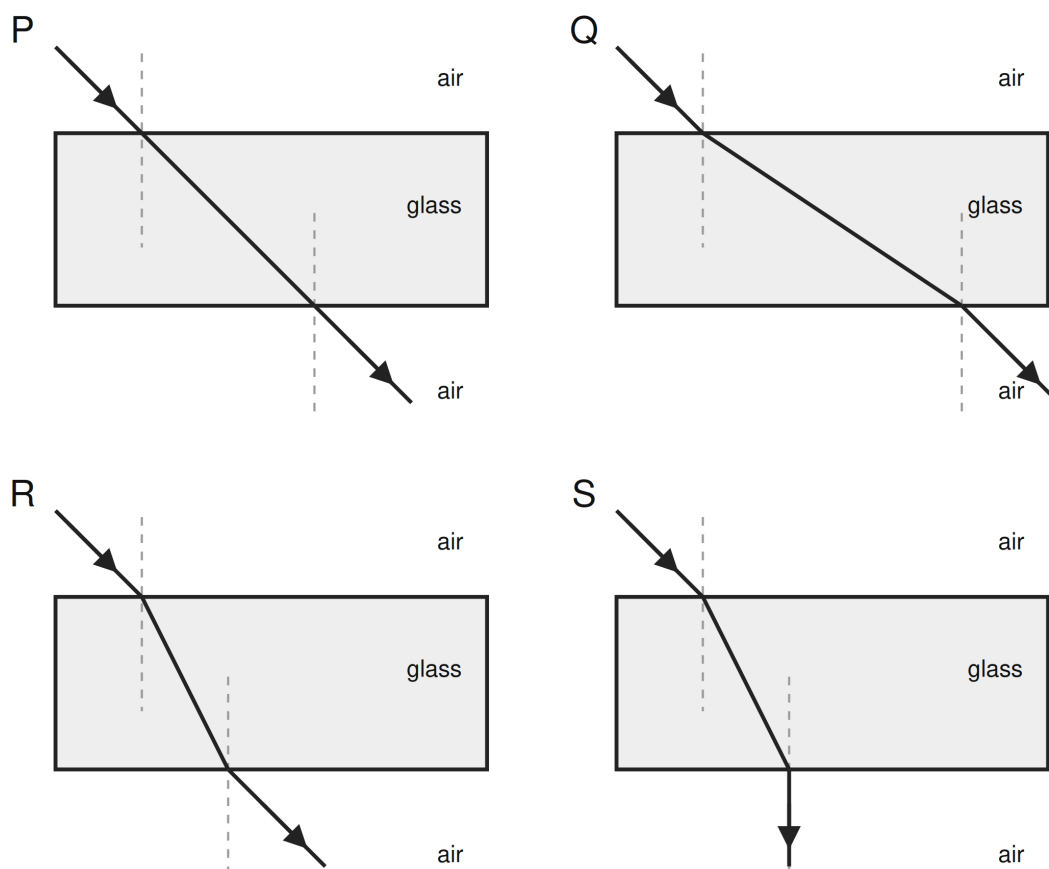
Which statement is correct?

- A** The observation proves that the quoted half-life is wrong.
- B** Exactly 36 of the remaining nuclei must decay during the next half-life.
- C** The 72 surviving nuclei are less likely to decay because they have already survived one half-life.
- D** The expected number remaining after the next half-life is 36, but the actual number may differ.
- E** The next decay must occur after exactly one seventy-second of a half-life.

38

Each diagram is intended to show a ray travelling from air through a rectangular glass block and back into air. Light travels more slowly in glass than in air. The two faces crossed are parallel.

Which diagram shows a possible path? Ignore partial reflection.



- A** P
- B** Q
- C** R
- D** S
- E** None of them

39

A sound wave travels to the right through a long tube of air. A tiny marked air particle is initially at P.

Which description of the particle's motion is correct as the wave passes?

- A** It travels to the right with the wave and stops far from P.
- B** It oscillates up and down, perpendicular to the tube.
- C** It moves back and forth along the tube, with no net transport over a complete oscillation.
- D** It remains stationary while the surrounding air carries the wave.
- E** It follows a circular path whose radius is the wavelength.

40

An LDR and a fixed $1.0\text{ k}\Omega$ resistor are connected in series across an ideal 9.0 V supply. At the fixed light intensity, the working LDR has resistance $2.0\text{ k}\Omega$.

A fault develops in exactly one component. An ideal voltmeter across the LDR now reads 9.0 V , and an ideal ammeter in series with the supply reads 4.5 mA . All connecting wires remain intact.

Which fault is consistent with both readings?

- A** The LDR has become an open circuit.
- B** The LDR has become a short circuit.
- C** The fixed resistor has become an open circuit.
- D** The fixed resistor has become a short circuit.
- E** The LDR resistance has increased to $3.0\text{ k}\Omega$.